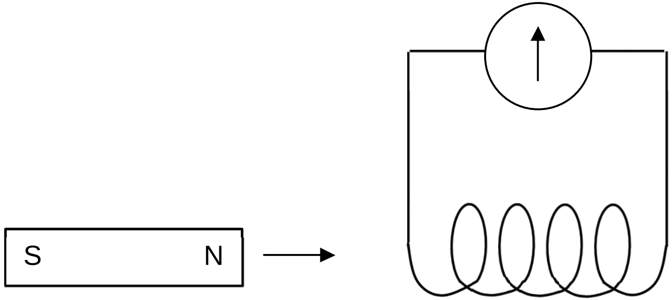
magnetic field, no electromagnetic force acts on the electron.

24 The North pole of a bar magnet is pushed into the end of a coil of wire. The maximum movement of the meter needle is 10 units to the left.



The South pole of the magnet is then pushed into the other end of the coil at half the speed.

What is the maximum movement of the meter needle?

A less than 10 units to the left

[Turn over

B less than 10 units to the right

C more than 10 units to the left

D more than 10 units to the right

Ans: A

Applying Lenz's law, a North will be induced on the left side of the coil when the North pole is pushed into the left end; A South pole will be induced on the right side when South is pushed into the right end. Polarity of the induced B-field in the coil is the same for both cases.

Since speed is halved, rate of change of magnetic flux linkage in the coil is less and a lower (e.m.f. and hence) current is induced.

25 The secondary coil of an ideal transformer delivers an r.m.s. current of 1.5 A to a load resistor